TOXION

Multitask Toxicity Prediction on Tox21

Fingerprint baseline • Graph fusion planned

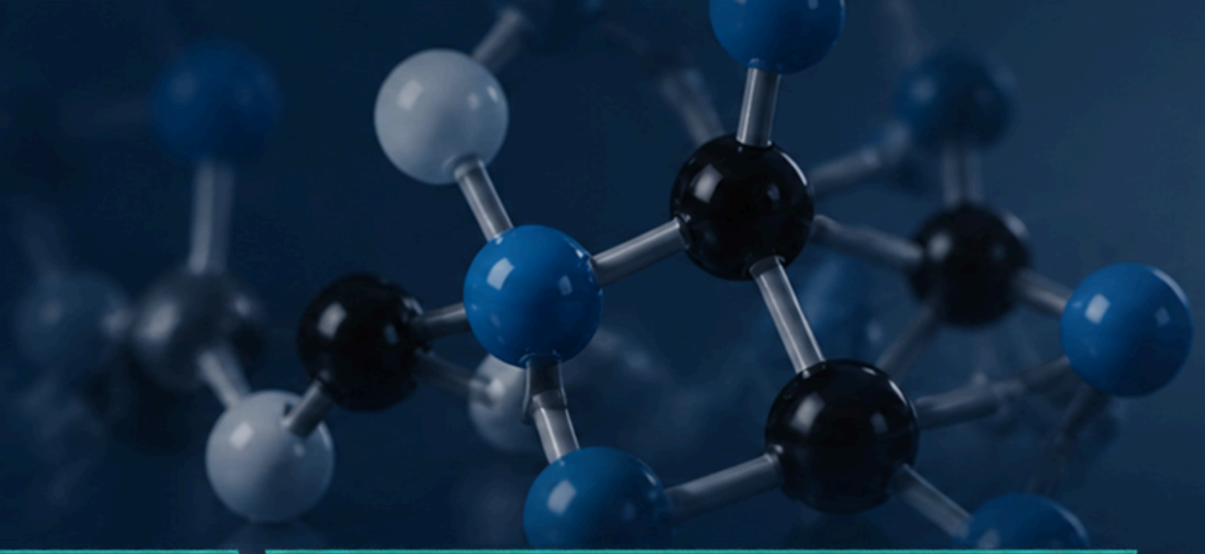
Problem

Predict 12 in-vitro assays, not human safety

17.1% missing labels; 2.9–16.2% actives

Similar chemical structures can **inflate** test results

Dataset



7,831
raw rows; 8 dropped

7,823
cleaned molecules

12
targets

7 nuclear receptor
5 stress response

80/10/10
scaffold split • seed 80

train | val | test

Current Work

01

Clean and explore data

RDKit cleaning; missing labels stay NaN

02

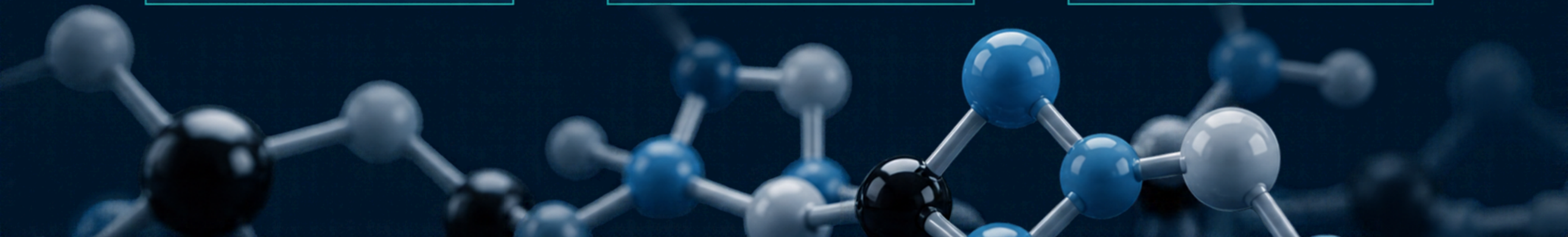
Analyze chemical space

815 PCA components explain 90% of variance

03

Train linear baseline

2048-bit Morgan fingerprints; 12 logistic outputs



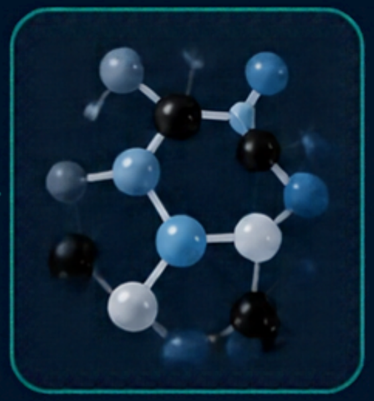
Planned Fusion

Next stage only — graph and fusion models are not yet evaluated

INPUT

SMILES string

Molecular structure



BRANCH 1

Message Passing Neural Network

(MPNN)

101100...101

010011...0110

2048 bits

BRANCH 2

Multilayer Perceptron

(MLP)

FEATURE FUSION

Graph embedding + encoded flat features

OUTPUT

12

assay predictions

Training

Masked cross-entropy;
ignore missing labels

Try 4 optimizers;
15 epochs each

Validation selects RMSprop
Learning rate: 0.001

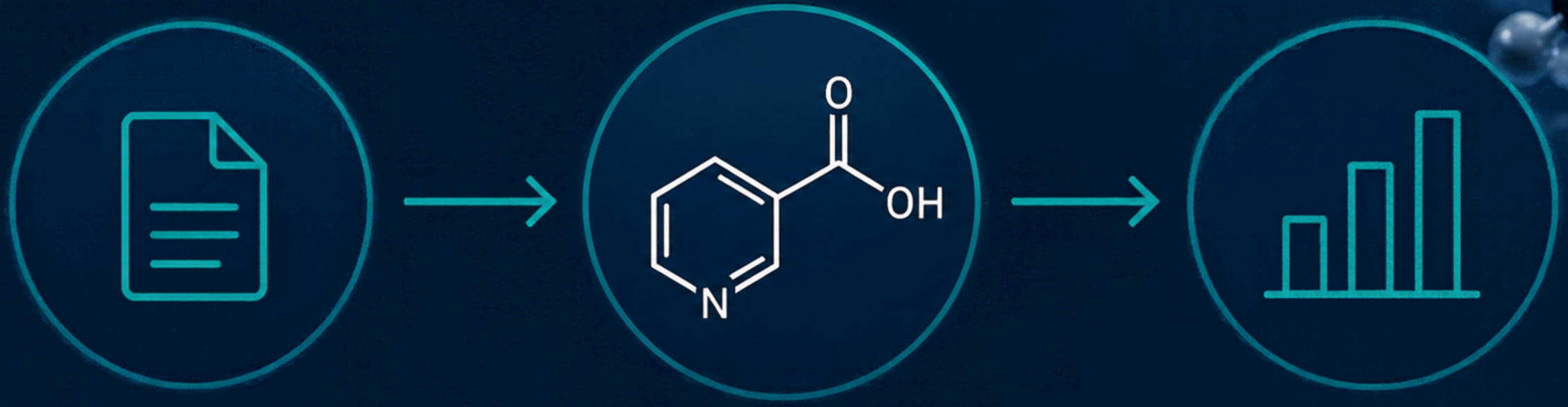
Best epoch: 28 of 43
Validation threshold: 0.25

Baseline Results

Metric	Linear baseline • One scaffold split, one seed
Train ROC-AUC	0.9023
Validation ROC-AUC	0.7274
Test ROC-AUC	0.6539
Test PR-AUC	**0.2883**

Test F1 • threshold 0.25 **0.2849**	Strong overfitting **Preliminary results**

Validated Pipeline



Clean and summarize data

Analyze fingerprints and scaffolds

Train and evaluate linear baseline

Conclusion

Fingerprint baseline overfits;
results are preliminary

Next: descriptors, alerts,
graph and fusion models

**Evaluate on 5 balanced
scaffold splits; calibrate**

Thank you / Questions